

OM RAJ

Agricultural Researcher • Environmental Analyst • Software Developer

📍 Patna, Bihar / Rupnagar, Punjab, India 📞 +91 7979734220 ✉ omraj.student@gmail.com 🌐 [omraj-website](#)

🆔 [ORCID: 0009-0002-7332-3962](#) [in linkedin.com/in/omraj](#)

👤 PROFESSIONAL SUMMARY

Interdisciplinary researcher, environmental analyst, and software developer with an **M.Sc. in Botany (Plant Pathology)** and hands-on project research experience at **IIT Ropar**. Uniquely blends wet-lab experimental rigor (spectrophotometry, chromatography, plant biochemical stress markers, wastewater testing) with full-stack desktop software engineering and digital agriculture knowledge systems. Proven record in executing government-sponsored research projects (PPCB and Punjab Excise Department), authoring peer-reviewed literature, developing production-ready scientific software deployed on the Microsoft Store, and designing digital public platforms. Equipped to drive impact across agri-tech knowledge engineering, environmental compliance monitoring, and computational research pipelines.

🔧 TECHNICAL & DOMAIN SKILLS MATRIX

Agriculture & Plant Pathology:

Crop disease diagnostics, pathogen identification, IDM, stress physiology, enzyme assays (SOD, CAT, POD, proline, MDA), biochar amendments, soil fertility profiling.

Air Quality & Instrumentation:

Aerosol Spectrophotometer, TVOC Analyzer, Carbon Analyzer, Formaldehyde Analyzer, sensor calibration, air pollution time-series spatiotemporal assessment.

Data Science, GIS & Automation:

MS Excel (advanced formulas, linear regression), Power BI interactive dashboards, QGIS / basic ArcGIS spatial mapping, satellite remote sensing interpretation.

Environmental & Water Engineering:

Wastewater testing (CETP, STP, ETP), river monitoring (pH, EC, TDS, TSS, BOD₅, COD, DO, alkalinity, coliforms, heavy metals) per APHA 24th Ed., BIS 10500, CPCB.

Software Development & Computing:

JavaScript, HTML5, CSS3, Electron.js, Node.js basics, REST API integration, Git/GitHub, Visual Studio Code.

Creative Design & Communication:

Adobe Photoshop, Canva Pro, Premiere Pro, CapCut, InShot, graphical abstracts for peer-reviewed journals, technical documentation, SOP authoring.

📁 PROFESSIONAL RESEARCH & WORK EXPERIENCE

Project Researcher – Environmental Monitoring & Plant–Soil Interactions

2025 – Present

Department of Civil Engineering, IIT Ropar | Supervisors: Dr. Indramani Dhada, Dr. Neelkanth Nirmalkar | PM: Dr. Harsh Sharma

- **Sustainable Agriculture & Biochar Project:** Investigated comparative impact of agricultural biomass feedstocks (pineapple peel, jackfruit peel, drek wood biochars) on soil physicochemical characteristics, nutrient retention, and crop physiological responses.
- **Plant Biochemical Assays:** Quantified defense enzyme kinetics (Superoxide Dismutase [SOD], Catalase [CAT], Peroxidase [POD], Proline, Malondialdehyde [MDA]) to evaluate plant resilience under abiotic and heavy-metal stress.
- **Wastewater & River Basin Monitoring (PPCB Project):** Conducted sampling and wet-chemical testing across CETPs, STPs, and Sutlej River basin; estimated BOD, COD, TSS, heavy metals, and coliforms.
- **Technical Environmental Audits (Punjab Excise Dept.):** Participated in audits of industrial distilleries, breweries, and bottling plants across Punjab evaluating Zero Liquid Discharge (ZLD) performance and compliance logs.

- **SOPs & Calibration:** Authored SOPs, calibrated precision sensor probes, and established linear regression curves for spectrophotometric assays.

Postgraduate Dissertation Researcher — River Water Quality & Runoff

2024 – 2025

Patna University | Supervisor: Prof. Dr. Meenakshi Singh (Head, Dept. of Biotechnology)

- Executed multi-point seasonal field sampling along major ghats of Ganga River at Patna. Analyzed physicochemical and biological parameters against WHO, CPCB, and BIS 10500 standards to assess agricultural runoff and urban drainage impact.

Medical Writing & Scientific Research Intern

Nov 2025

YuvaIntern (in association with NSDC & Henry Harvin)

- Executed systematic literature reviews across PubMed, Scopus, Web of Science; synthesized biomedical and environmental research into technical summaries with APA, IEEE, and Vancouver citations.

Freelance Creative Designer & Digital Media Specialist

2021 – Present

Freelance & MARKAD LIMITED

- Designed graphical abstracts for journal submissions, academic branding, and video explainers. Directed media campaigns for Patna University Student Union Elections (2022 & 2025).

 EDUCATION & ACADEMIC QUALIFICATIONS

Degree / Program	Specialization / Discipline	Institution / University	Year	Score / CGPA
M.Sc. in Botany	Plant Pathology	Department of Botany, Patna University	2025	7.56 CGPA
B.Sc. in Botany	Plant Sciences & Chemistry	Patna Science College, Patna University	2023	63.50%
Diploma in Computer Applications (DCA)	Software & Computing Systems	Vinita Computer Education	2022	64.40%
Senior Secondary (Class XII, Science)	Physics, Chemistry, Biology (PCB)	Shanti Niketan Sr. Sec. School (CBSE)	2019	58.40%
Secondary School (Class X)	General Science & Mathematics	Sunshine Prep. / High School (CBSE)	2017	81.70%

 SOFTWARE ENGINEERING & PRODUCTION INNOVATIONS

- AWE LAB Academic Portal (IIT Ropar):** Architected, designed, and deployed official laboratory web portal for Dr. Indramani Dhada's team at IIT Ropar. ([AWE LAB Portal](#))
- SNIFF DOI — Academic Literature Discovery Engine (Microsoft Store):** Desktop software indexing 250M+ articles. Features 6–tier research gap synthesis and CSL engine (APA, IEEE, Harvard, Vancouver). ([Microsoft Store](#))
- WASH (Water Analysis Suite Hub) — Microsoft Store:** Environmental compliance engine automating APHA 24th Ed & BIS 10500 algorithms, spectrophotometer linear regression calibration, and NABL PDF reports. ([Microsoft Store](#))
- Calculator Luxe & Clock Span (Watch Span):** High-precision scientific desktop calculator & lab reaction kinetics timer published on Microsoft Store.

 PUBLICATIONS & RESEARCH MANUSCRIPTS

- Peer-Reviewed Journal Article:** Raj, O., & Ambreen, D. (2026). Assessment of Physicochemical Parameters of Water Samples from the Ganga River at Patna. *International Journal of Versatile Research and Analysis (IJVRA)*, 3(4), 784–810. ISSN: 2984–8903. Paper ID: IJVRA2603368.
- Research Preprint (Air Quality):** Raj, O., & Dhada, I. Spatiotemporal Assessment of Ambient Air Quality in Selected Urban, Tourist and Industrial Regions of Himachal Pradesh, India: A Case Study. *Research Square*. DOI: [10.21203/rs.3.rs-10492149/v1](#)
- Manuscript in Preparation (Sustainable Agriculture):** Raj, O., & Dhada, I. Comparative Assessment of Pineapple Peel, Jackfruit Peel, and Drek Wood Biochar on Plant Growth, Soil Fertility, and Biochemical Responses. (Evaluates SOD, CAT, POD, proline, MDA kinetics).

 WORKSHOPS & PROFESSIONAL CERTIFICATIONS

- Fundamentals of Remote Sensing:** NASA "Train – Empower – Advance" Initiative (Sept 2026) – Satellite observations & spatial datasets.
- Advances in Genome Biology & Gene Regulation:** School of Life Sciences, JNU, New Delhi (June 2025).
- Marine Data 4 Asia 2025:** Copernicus Marine Service & Mercator Ocean International, EU Program (June 2025).
- Power BI & AI Interactive Dashboards:** OfficeMaster (Sept 2025).
- Cancer Genomics, Clinical Research & AI:** Biotechnika (Sept 2025).

LEADERSHIP & EXTENSION ACTIVITIES

- **Event & Sports Coordinator:** Patna Science College (2023, 2024, 2025). Managed logistics, budgeting, and operations for 3 consecutive Annual Sports Meets. Recipient of *The Catalyst Award (2025)*.
- **Volunteer Coordinator:** Patna University Alumni Meet (2023, 2025). Managed guest reception, VIP protocol, and stage management.

PROFESSIONAL REFERENCES

Dr. Indramani Dhada

Assistant Professor, Dept. of Civil Engineering
Indian Institute of Technology (IIT) Ropar, Rupnagar –
140001, Punjab, India
Email: indhada@iitrpr.ac.in

Dr. Raj Laxmi Singh

Assistant Professor, Dept. of Botany
Patna University, Patna – 800005, Bihar, India
Email: rajlaxmi1982singh@gmail.com